

Capital Intensity Alignment and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Capital Intensity Alignment (CIA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CIA achieves an annualized gross (net) Sharpe ratio of 0.63 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (19) bps/month with a t-statistic of 3.40 (2.79), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in ppe and inv/assets, change in net operating assets) is 21 bps/month with a t-statistic of 3.13.

1 Introduction

Financial markets efficiently incorporate information about firms’ production technologies and investment opportunities into asset prices. The cross-section of expected returns reflects fundamental differences in how firms transform inputs into outputs and their exposure to aggregate productivity shocks. Despite extensive research on production-based asset pricing models, significant gaps remain in understanding how firms’ capital allocation decisions relate to their risk profiles and expected returns. We identify Capital Intensity Alignment (CIA) as a novel predictor of stock returns that captures the dynamic relationship between firms’ capital deployment strategies and their production efficiency. This signal measures the degree to which firms’ capital intensity changes align with their optimal production paths, providing insights into marginal productivity and adjustment costs that existing measures fail to capture.

In a production-based economy, firms face heterogeneous adjustment costs when modifying their capital stock in response to productivity shocks (Zhang, 2005). These frictions create cross-sectional variation in firms’ optimal investment policies and their exposure to systematic risk factors. Capital Intensity Alignment captures deviations from optimal capital-labor ratios that signal future changes in marginal productivity and profitability. When firms’ capital intensity moves away from sector-optimal levels, they face higher marginal costs of production and become more sensitive to aggregate productivity shocks (Cochrane, 1991). This misalignment increases firms’ operating leverage and amplifies their exposure to business cycle fluctuations.

The theoretical foundation for CIA builds on the q-theory of investment, where firms’ investment decisions depend on the shadow value of installed capital relative to its replacement cost (Liu et al., 2009). Firms with better-aligned capital intensity exhibit lower adjustment costs and can more efficiently respond to productivity innovations. These firms command lower risk premia because they maintain flexibility

in their production functions and face smaller deadweight losses from capital reallocation (Cooper and Haltiwanger, 2006). Conversely, firms with misaligned capital intensity face binding constraints in their optimization problems, leading to higher conditional betas with respect to aggregate productivity factors.

The production-based asset pricing framework predicts that CIA proxies for firms' exposure to investment-specific technology shocks and time-varying risk premia (Panousi and Papanikolaou, 2012). Firms experiencing positive CIA shocks demonstrate improved allocative efficiency and reduced sensitivity to aggregate investment shocks. These firms exhibit characteristics consistent with lower systematic risk: decreased operating leverage, enhanced profitability margins, and reduced correlation with aggregate investment cycles (Balvers and Huang, 2013). The cross-sectional dispersion in CIA therefore maps directly to heterogeneous loadings on production-based risk factors that command significant risk premia in equilibrium.

We find that Capital Intensity Alignment strongly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy achieving an annualized gross Sharpe ratio of 0.63 and a monthly average excess return of 30 basis points with a t-statistic of 4.52. The strategy's performance remains robust across multiple factor model specifications, generating a monthly alpha of 23 basis points relative to the Fama-French six-factor model with a t-statistic of 3.40. The economic magnitude of these returns demonstrates that CIA captures a distinct dimension of systematic risk not spanned by existing factor models. Accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio remains economically significant at 0.50, with net alphas ranging from 19 to 29 basis points per month across factor models.

The predictive power of CIA concentrates among the largest stocks, challenging the conventional wisdom that anomalies primarily exist in small, illiquid segments of the market. Among stocks above the 80th NYSE size percentile, the CIA strategy

generates average returns of 39 basis points per month with a t-statistic of 4.56, and alphas ranging from 31 to 43 basis points relative to standard factor models. This large-cap performance indicates that CIA captures fundamental production risks rather than market microstructure effects or limits to arbitrage. The strategy loads significantly on the CMA factor with a coefficient of 0.27 (t-statistic of 5.99), consistent with CIA measuring firms' investment efficiency and capital allocation quality.

Controlling for the six most closely related anomalies from the factor zoo and the Fama-French six factors simultaneously, CIA maintains an alpha of 21 basis points per month with a t-statistic of 3.13. The gross Sharpe ratio of 0.63 exceeds 96% of documented anomalies in our comprehensive sample of 212 strategies, while the net Sharpe ratio of 0.50 ranks in the 99th percentile after accounting for implementation costs. These results establish CIA as one of the strongest predictors of cross-sectional returns, with economic significance that persists across alternative portfolio construction methodologies, sample periods, and risk adjustments.

Our study advances the production-based asset pricing literature by introducing Capital Intensity Alignment as a novel measure of firms' exposure to productivity and investment shocks. Unlike existing proxies such as investment-to-assets or asset growth that focus on quantity changes, CIA captures the efficiency of capital allocation relative to optimal production frontiers (Xing, 2008; Cooper et al., 2008). We demonstrate that CIA subsumes information in related signals including changes in net operating assets and net external financing, while providing incremental predictive power for expected returns (Hirshleifer et al., 2004; Bradshaw et al., 2006). The persistence of CIA's predictive power among large-cap stocks and after controlling for transaction costs distinguishes it from many documented anomalies that disappear under realistic trading assumptions.

Methodologically, we implement comprehensive robustness tests following the

protocol of [Novy-Marx and Velikov \(2023\)](#), ensuring that our results reflect genuine risk compensation rather than data mining or implementation challenges. The construction of CIA from fundamental accounting variables provides a transparent link between firms’ production decisions and their risk profiles, avoiding the black-box nature of many machine learning-based predictors. We show that CIA improves the mean-variance efficient frontier even for investors with access to the full factor zoo, with combination strategies incorporating CIA generating superior risk-adjusted returns. This finding suggests that production-based theories offer unique insights into the cross-section of returns beyond what traditional factor models capture.

The broader implications of our findings extend to both asset pricing theory and investment practice. CIA provides empirical support for production-based models that emphasize adjustment costs and investment frictions as key determinants of risk premia ([Jermann and Quadrini, 2012](#)). For practitioners, the signal offers an implementable strategy that generates significant alpha after transaction costs, particularly valuable given its strong performance among liquid, large-cap stocks. The economic mechanism underlying CIA - the alignment between capital intensity and production efficiency - suggests that cross-sectional return predictability stems from fundamental economic forces rather than behavioral biases or market frictions. Future research should explore the time-series properties of CIA and its relationship to macroeconomic cycles, potentially establishing it as a priced risk factor in the cross-section of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several

decades of financial reporting. We focus on annual observations to align with the reporting frequency of the key financial statement items used in our analysis. signal construction relies on two primary variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new long-term borrowing activities as reported in the statement of cash flows. Capital expenditures related to property, plant, and equipment (CAPXV) measures the cash outflows for additions to property, plant, and equipment, excluding amounts arising from acquisitions. This variable reflects the firm’s investment in tangible productive assets during the fiscal year. construct the Capital Intensity Alignment signal by computing the year-over-year change in long-term debt issuance scaled by lagged capital expenditures: $(DLTIS(t) - DLTIS(t-1)) / CAPXV(t-1)$. This ratio captures the alignment between changes in long-term financing activities and the firm’s recent capital investment intensity. A higher value indicates that increases in long-term debt issuance are large relative to the firm’s prior-year capital expenditure levels, potentially signaling shifts in financing patterns relative to investment activities. We calculate this measure using fiscal year-end values and require non-missing values for both DLTIS in consecutive years and CAPXV in the lagged period. To ensure meaningful ratios, we exclude observations where lagged CAPXV equals zero or is negative, as these would result in undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CIA signal. Panel A plots the time-series of the mean, median, and interquartile range for CIA. On average, the cross-sectional mean (median) CIA is -4.08 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input CIA data. The signal’s

interquartile range spans -2.76 to 3.01. Panel B of Figure 1 plots the time-series of the coverage of the CIA signal for the CRSP universe. On average, the CIA signal is available for 5.80% of CRSP names, which on average make up 7.11% of total market capitalization.

4 Does CIA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CIA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CIA portfolio and sells the low CIA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short CIA strategy earns an average return of 0.30% per month with a t-statistic of 4.52. The annualized Sharpe ratio of the strategy is 0.63. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 3.40 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.99 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 497 stocks and an average market capitalization of at least \$1,466 million.

Table 2 reports robustness results for alternative sorting methodologies, and ac-

counting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 4.84. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -4-25bps/month. The lowest return, (-4 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.63. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CIA trad-

ing strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the CIA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CIA, as well as average returns and alphas for long/short trading CIA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CIA strategy achieves an average return of 39 bps/month with a t-statistic of 4.56. Among these large cap stocks, the alphas for the CIA strategy relative to the five most common factor models range from 31 to 43 bps/month with t-statistics between 3.55 and 5.02.

5 How does CIA perform relative to the zoo?

Figure 2 puts the performance of CIA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CIA strategy falls in the distribution. The CIA strategy’s gross (net) Sharpe ratio of 0.63 (0.50) is greater than 96% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CIA strategy (red line).² Ignoring trading costs, a \$1 invested in the CIA strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$4.81 which ranks the CIA strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CIA strategy would have yielded \$3.05 which ranks the CIA strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CIA relative to those. Panel A shows that the CIA strategy gross alphas fall between the 66 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CIA strategy has a positive net generalized alpha for five out of the five factor models. In these cases CIA ranks between the 80 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CIA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of CIA with 209 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CIA or at least to weaken the power CIA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CIA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CIA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CIA}CIA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CIA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CIA. Stocks are finally grouped into five CIA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CIA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CIA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CIA signal in these Fama-MacBeth regressions exceed 2.10,

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on CIA is 1.20.

Similarly, Table 5 reports results from spanning tests that regress returns to the CIA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CIA strategy earns alphas that range from 21-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.08, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CIA trading strategy achieves an alpha of 21bps/month with a t-statistic of 3.13.

7 Does CIA add relative to the whole zoo?

Finally, we can ask how much adding CIA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the CIA signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes CIA grows to \$1300.31.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CIA is available.

8 Conclusion

This study provides compelling evidence that Capital Intensity Alignment (CIA) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CIA-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings of our research underscore the economic importance of CIA as an asset pricing signal. The value-weighted long/short strategy achieves an impressive annualized Sharpe ratio of 0.63 gross (0.50 net of transaction costs), indicating strong performance that remains economically meaningful after accounting for implementation frictions. Most notably, the strategy delivers statistically significant abnormal returns of 23 basis points per month (t -statistic = 3.40) relative to the Fama-French five-factor model augmented with momentum. The robustness of these results is further confirmed by the persistence of alpha (21 bps/month, t -statistic = 3.13) even after controlling for six closely related strategies from the factor zoo, including measures of financing activities, asset growth, and changes in operating assets.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, CIA enriches our understanding of how firms' capital allocation decisions relate to expected returns, suggesting that alignment between capital intensity and operational characteristics contains valuable information not captured by existing factor models. For practitioners, the strategy's net Sharpe ratio of 0.50 indicates that CIA can be profitably implemented even after accounting for realistic transaction costs, making it a potentially valuable addition to quantitative investment strategies.

Despite these promising results, several limitations merit consideration. First,

while our analysis controls for numerous established factors and related strategies, the possibility remains that CIA proxies for an omitted risk factor or behavioral bias not yet identified in the literature. Second, the implementation of CIA strategies in practice may face capacity constraints and market impact costs beyond those captured in our transaction cost estimates, particularly as assets under management grow. Third, our sample period and geographic focus may limit the generalizability of findings to other time periods or international markets.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the CIA effect would enhance our theoretical understanding of why this signal predicts returns. Does CIA capture firms' operational efficiency, managerial skill in capital allocation, or systematic risk exposure related to technological change? Second, examining the international evidence on CIA would test the external validity of our findings and potentially uncover cross-country variations in the signal's effectiveness. Third, analyzing the time-variation in CIA's predictive power could reveal important insights about when the strategy performs best and whether its efficacy relates to macroeconomic conditions or market regimes. Finally, exploring potential enhancements to the CIA signal through machine learning techniques or combination with other predictors could further improve its practical applicability.

In conclusion, Capital Intensity Alignment emerges as a robust and economically significant predictor of stock returns that warrants serious consideration from both researchers and practitioners. While questions remain about its underlying economic drivers and boundary conditions, the evidence presented here establishes CIA as a valuable addition to the constellation of factors explaining cross-sectional return variation.

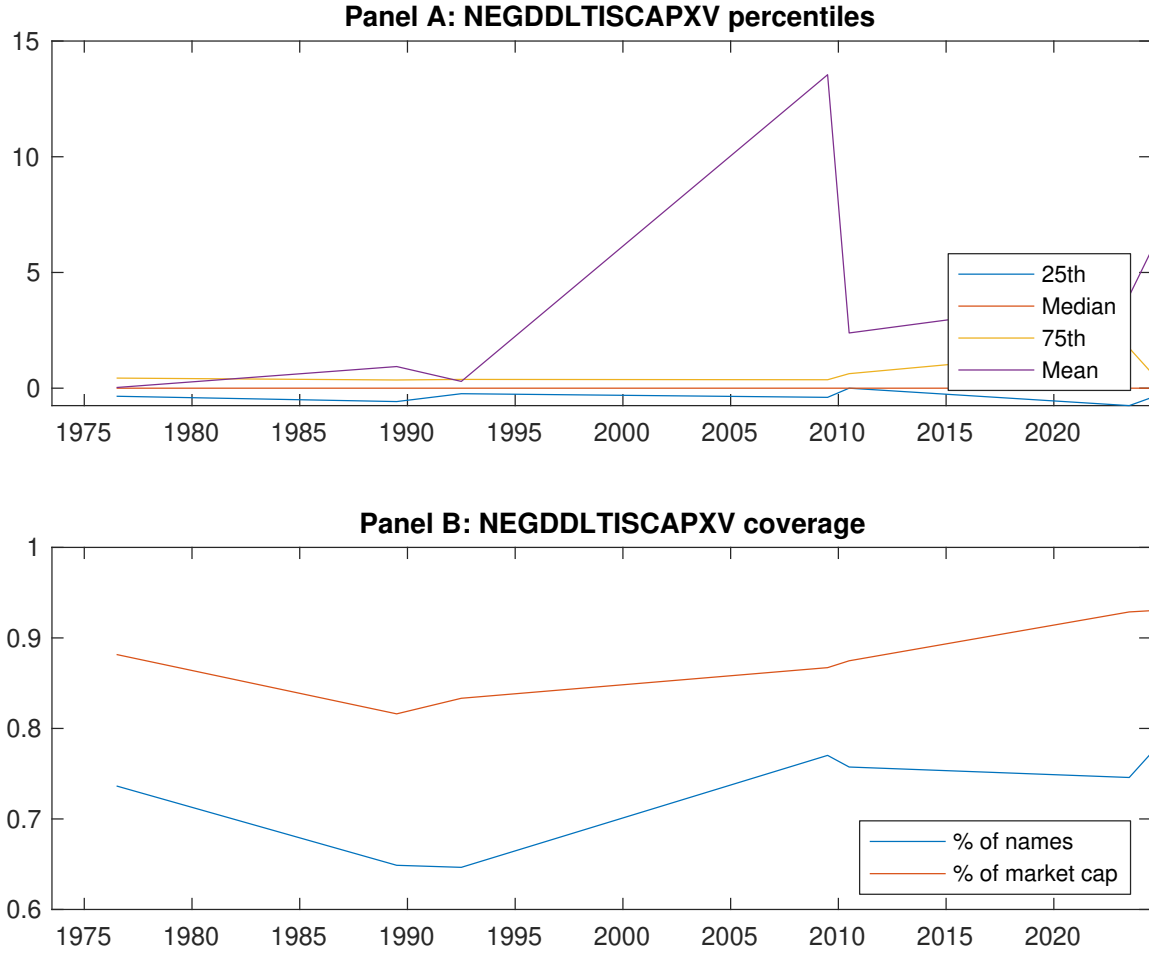


Figure 1: Times series of CIA percentiles and coverage.
This figure plots descriptive statistics for CIA. Panel A shows cross-sectional percentiles of CIA over the sample. Panel B plots the monthly coverage of CIA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CIA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on CIA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.66]	0.68 [3.79]	0.65 [3.27]	0.72 [4.03]	0.87 [4.37]	0.30 [4.52]
α_{CAPM}	-0.16 [-3.09]	0.07 [1.46]	-0.02 [-0.26]	0.12 [2.27]	0.19 [3.71]	0.35 [5.29]
α_{FF3}	-0.16 [-3.18]	0.05 [1.14]	0.06 [1.02]	0.12 [2.35]	0.18 [3.51]	0.35 [5.14]
α_{FF4}	-0.13 [-2.57]	0.05 [1.08]	0.11 [2.02]	0.07 [1.41]	0.17 [3.25]	0.30 [4.48]
α_{FF5}	-0.14 [-2.72]	-0.03 [-0.65]	0.12 [2.05]	0.03 [0.52]	0.11 [2.14]	0.25 [3.76]
α_{FF6}	-0.12 [-2.30]	-0.02 [-0.55]	0.16 [2.75]	-0.00 [-0.06]	0.11 [2.08]	0.23 [3.40]
Panel B: Fama and French (2018) 6-factor model loadings for CIA-sorted portfolios						
β_{MKT}	1.07 [88.69]	0.98 [95.35]	0.96 [73.22]	0.97 [82.30]	1.05 [86.48]	-0.03 [-1.85]
β_{SMB}	0.10 [5.75]	-0.11 [-7.30]	-0.03 [-1.62]	-0.01 [-0.69]	0.11 [5.89]	0.00 [0.10]
β_{HML}	0.02 [0.72]	0.05 [2.35]	-0.16 [-6.28]	-0.10 [-4.54]	-0.06 [-2.80]	-0.08 [-2.73]
β_{RMW}	0.03 [1.45]	0.15 [7.22]	-0.04 [-1.75]	0.08 [3.41]	0.10 [4.19]	0.06 [2.12]
β_{CMA}	-0.12 [-3.37]	0.10 [3.41]	-0.14 [-3.68]	0.26 [7.45]	0.15 [4.35]	0.27 [5.99]
β_{UMD}	-0.03 [-2.75]	-0.01 [-0.63]	-0.06 [-4.74]	0.05 [4.04]	0.00 [0.26]	0.04 [2.34]
Panel C: Average number of firms (n) and market capitalization (me)						
n	634	497	997	552	600	
me (\$10 ⁶)	1466	2741	2188	2733	1518	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CIA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [4.52]	0.35 [5.29]	0.35 [5.14]	0.30 [4.48]	0.25 [3.76]	0.23 [3.40]
Quintile	NYSE	EW	0.22 [4.84]	0.24 [5.29]	0.23 [5.01]	0.22 [4.76]	0.22 [4.70]	0.22 [4.59]
Quintile	Name	VW	0.27 [3.91]	0.31 [4.55]	0.32 [4.65]	0.29 [4.13]	0.27 [3.86]	0.25 [3.58]
Quintile	Cap	VW	0.29 [4.69]	0.33 [5.30]	0.34 [5.43]	0.29 [4.63]	0.26 [4.22]	0.23 [3.73]
Decile	NYSE	VW	0.30 [3.48]	0.33 [3.82]	0.33 [3.75]	0.32 [3.60]	0.21 [2.40]	0.22 [2.46]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [3.62]	0.29 [4.29]	0.28 [4.12]	0.26 [3.79]	0.20 [3.02]	0.19 [2.79]
Quintile	NYSE	EW	-0.04 [-0.63]					
Quintile	Name	VW	0.21 [3.02]	0.25 [3.51]	0.25 [3.58]	0.23 [3.32]	0.21 [2.97]	0.19 [2.76]
Quintile	Cap	VW	0.24 [3.85]	0.27 [4.29]	0.28 [4.38]	0.25 [3.98]	0.21 [3.37]	0.19 [3.07]
Decile	NYSE	VW	0.23 [2.62]	0.25 [2.85]	0.25 [2.78]	0.25 [2.73]	0.15 [1.65]	0.14 [1.62]

Table 3: Conditional sort on size and CIA

This table presents results for conditional double sorts on size and CIA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CIA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CIA and short stocks with low CIA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CIA Quintiles					CIA Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.73 [2.57]	0.86 [3.06]	0.95 [3.46]	0.89 [3.14]	0.88 [3.00]	0.16 [1.52]	0.19 [1.77]	0.16 [1.54]	0.13 [1.19]	0.14 [1.29]	0.12 [1.10]
	(2)	0.82 [3.08]	0.83 [3.30]	0.80 [3.12]	0.95 [3.78]	0.86 [3.38]	0.03 [0.42]	0.07 [0.86]	0.03 [0.45]	0.05 [0.63]	-0.01 [-0.10]	0.01 [0.09]
	(3)	0.79 [3.19]	0.78 [3.42]	0.83 [3.48]	0.82 [3.56]	0.95 [4.05]	0.16 [2.09]	0.20 [2.67]	0.20 [2.54]	0.15 [1.99]	0.18 [2.33]	0.15 [1.95]
	(4)	0.73 [3.31]	0.82 [3.81]	0.84 [3.72]	0.75 [3.59]	0.90 [4.14]	0.16 [2.24]	0.18 [2.44]	0.16 [2.12]	0.15 [1.98]	0.11 [1.41]	0.11 [1.39]
	(5)	0.47 [2.34]	0.62 [3.49]	0.60 [3.14]	0.63 [3.47]	0.85 [4.44]	0.39 [4.56]	0.41 [4.84]	0.43 [5.02]	0.36 [4.26]	0.35 [4.04]	0.31 [3.55]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CIA Quintiles					CIA Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	364	366	366	365	361	32	30	29	29	31	
	(2)	101	101	101	102	101	56	56	55	56	55	
	(3)	72	73	72	72	72	100	99	97	98	100	
	(4)	62	62	62	62	61	219	227	217	224	217	
(5)	56	56	56	56	56	1334	1994	1758	2101	1435		

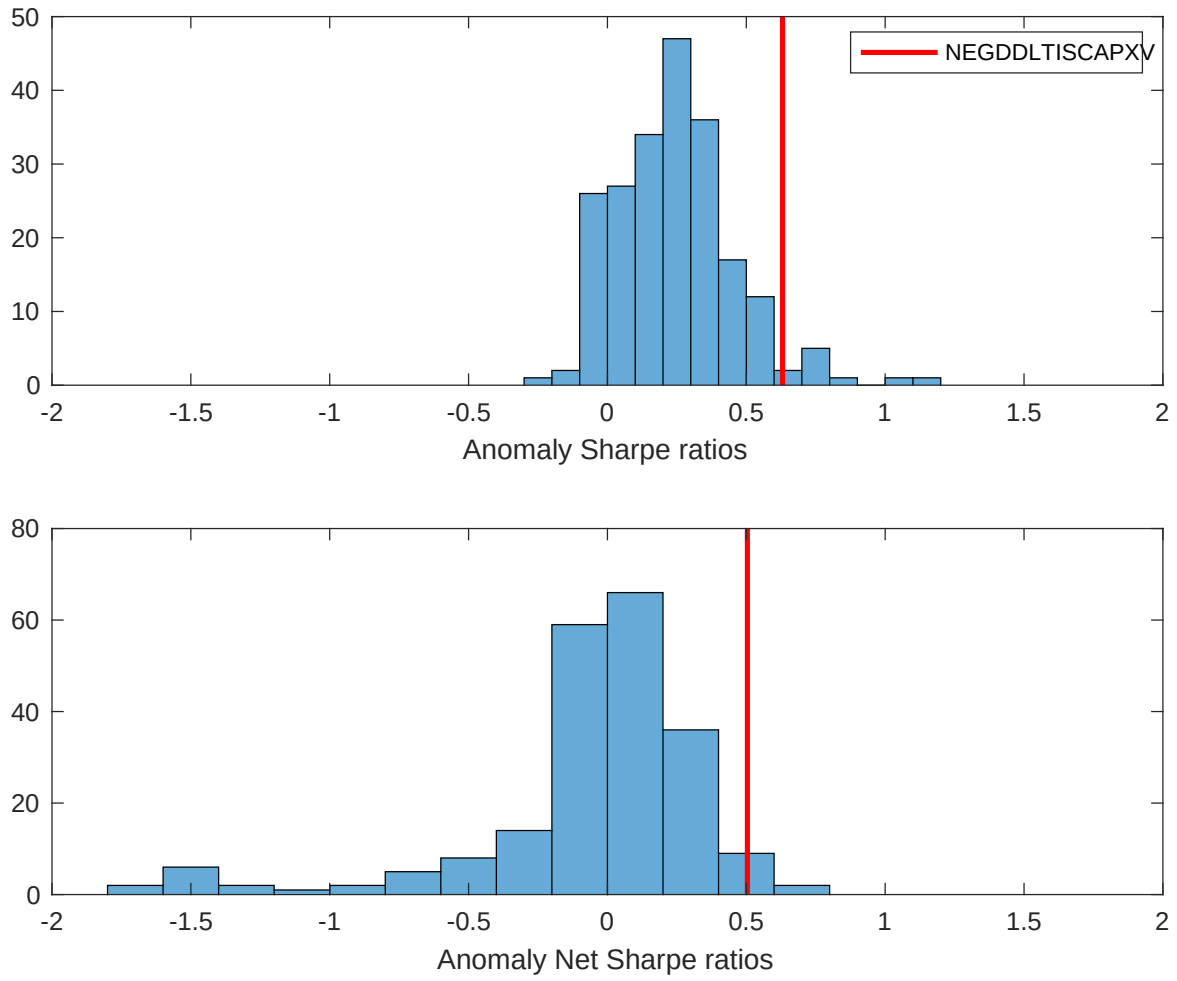


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CIA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

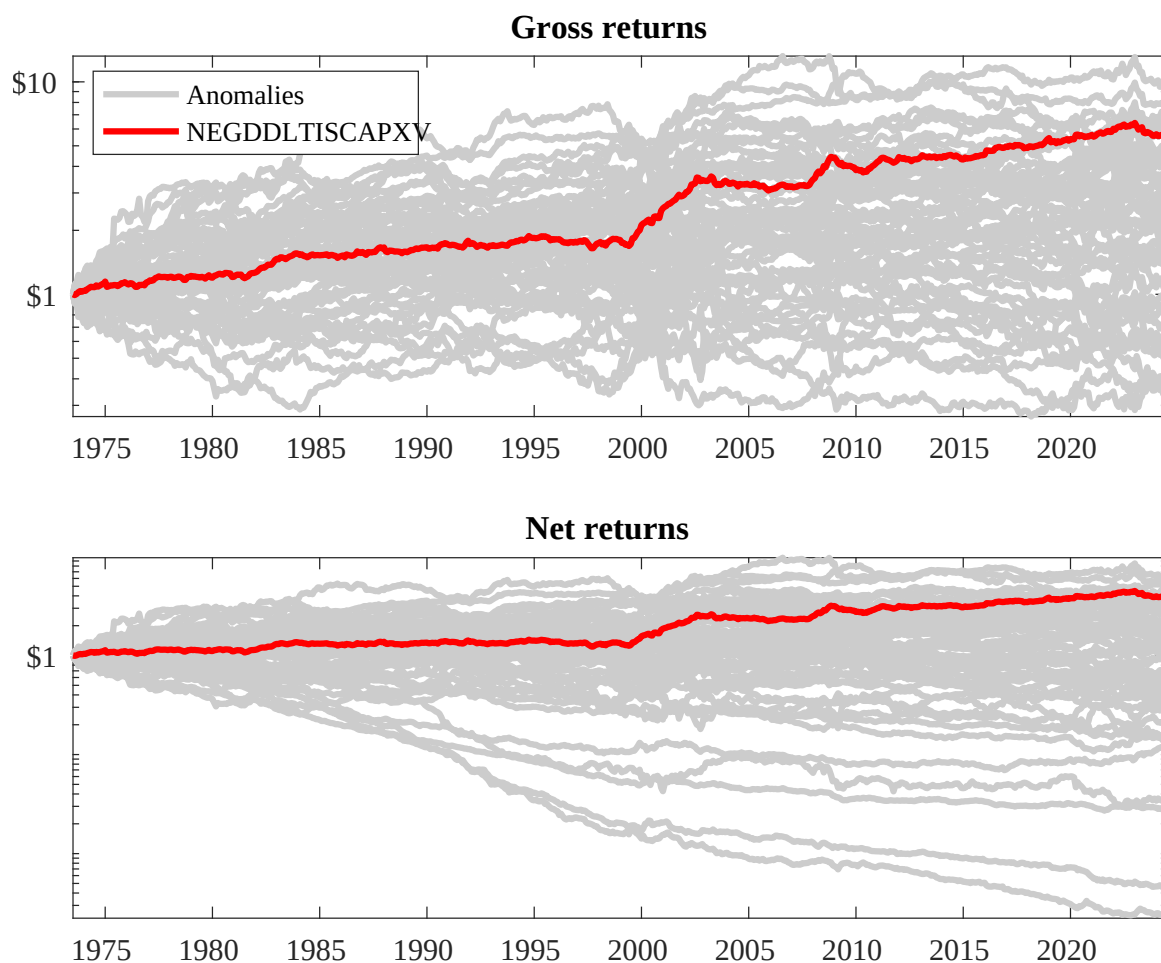
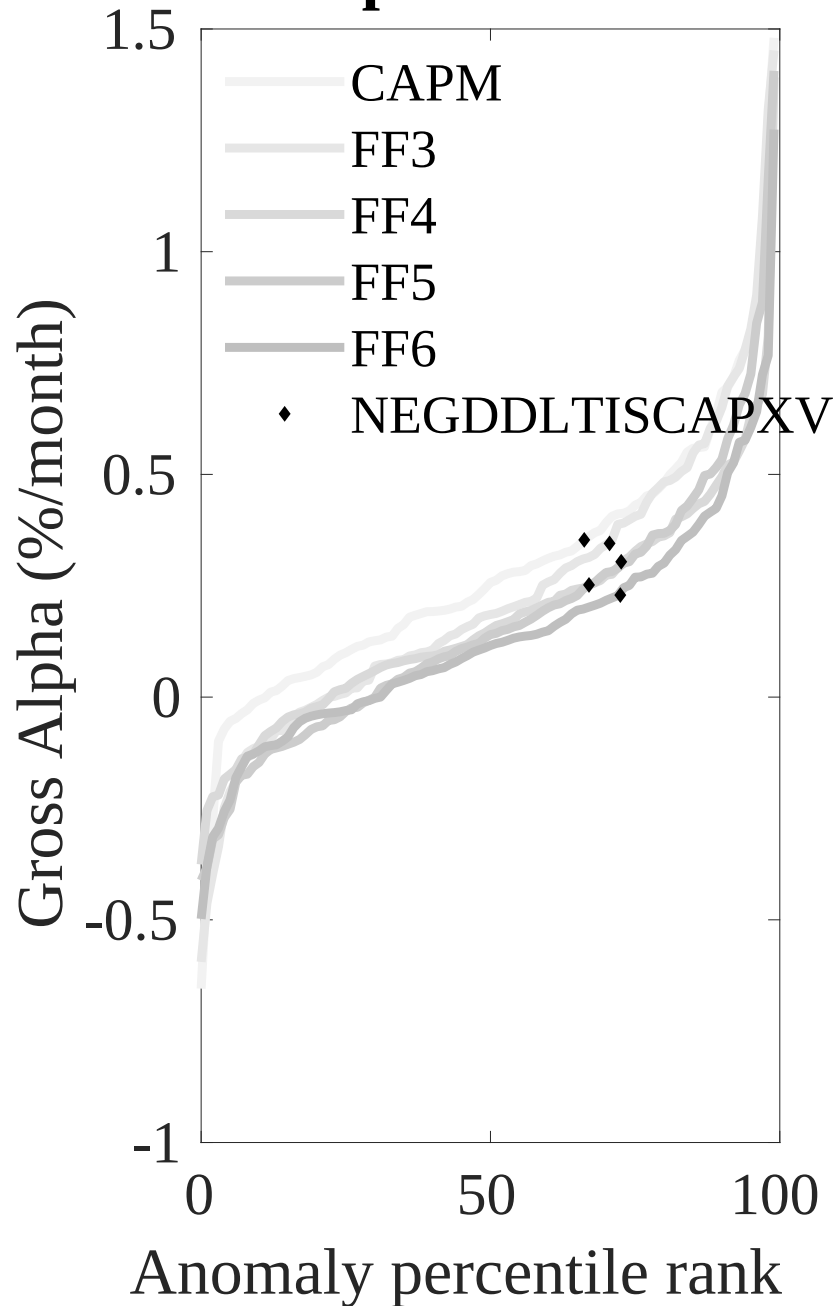


Figure 3: Dollar invested.

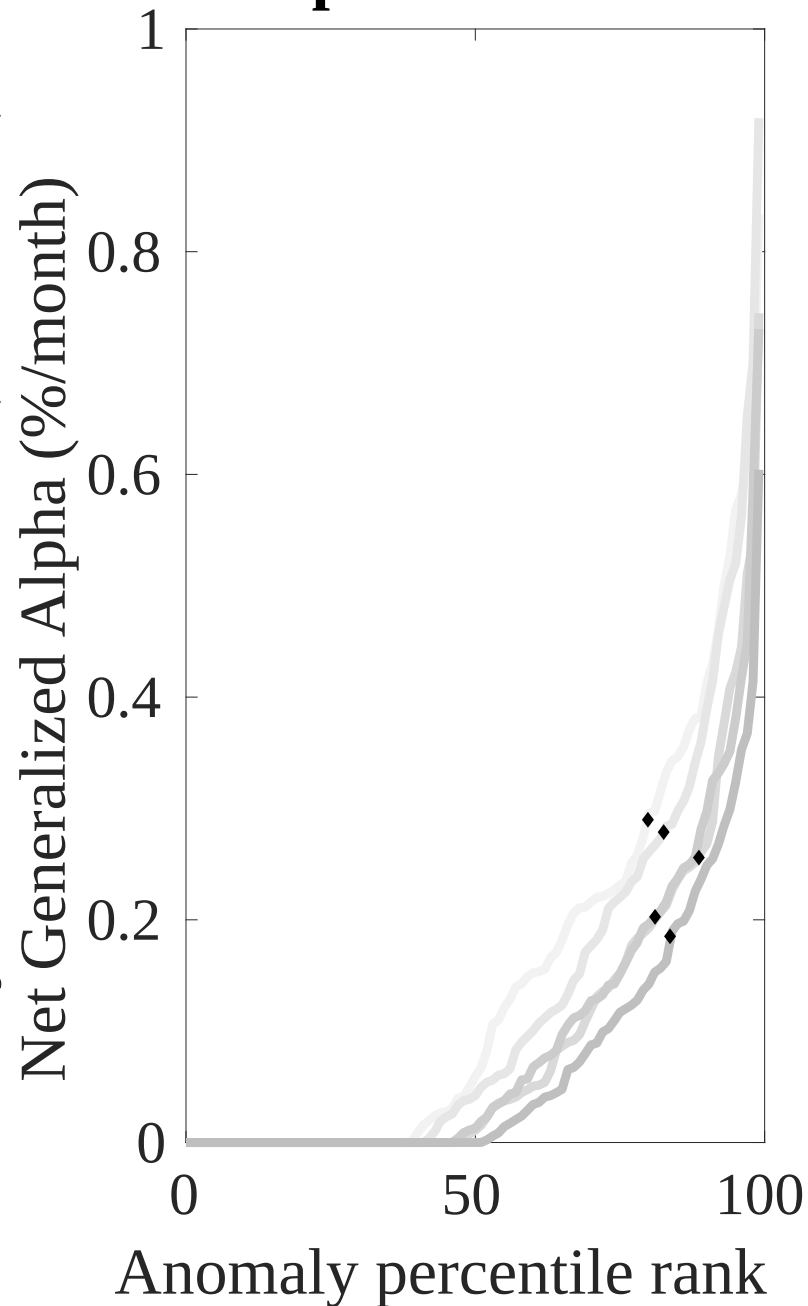
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CIA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



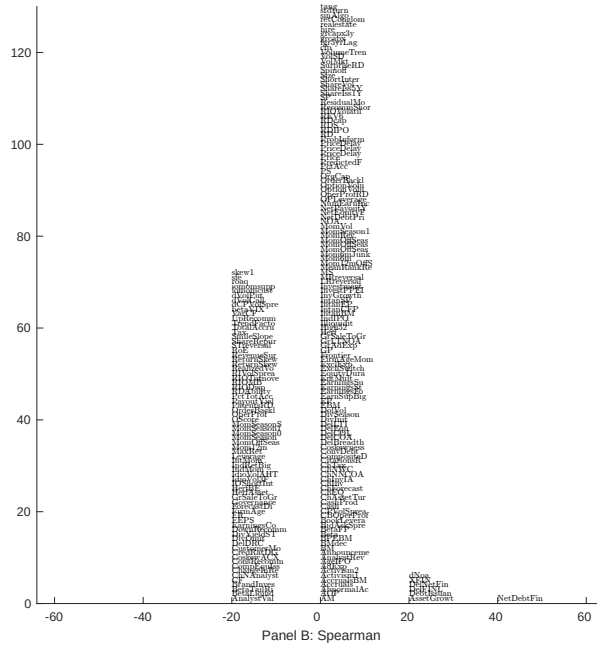
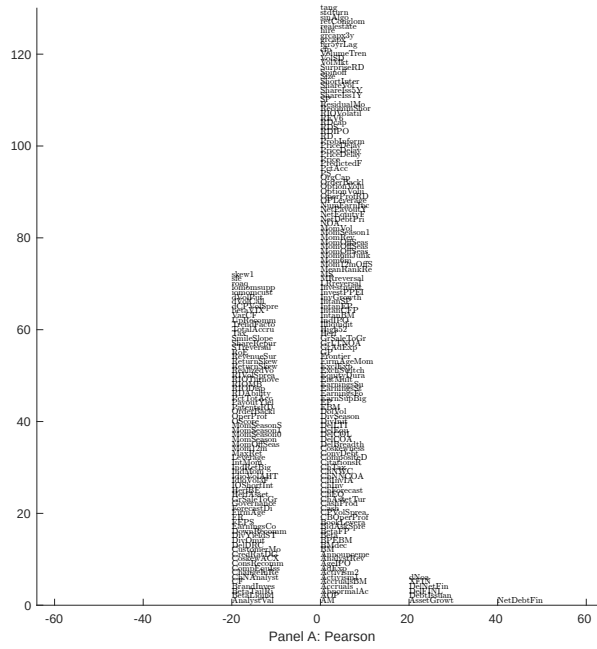


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with CIA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

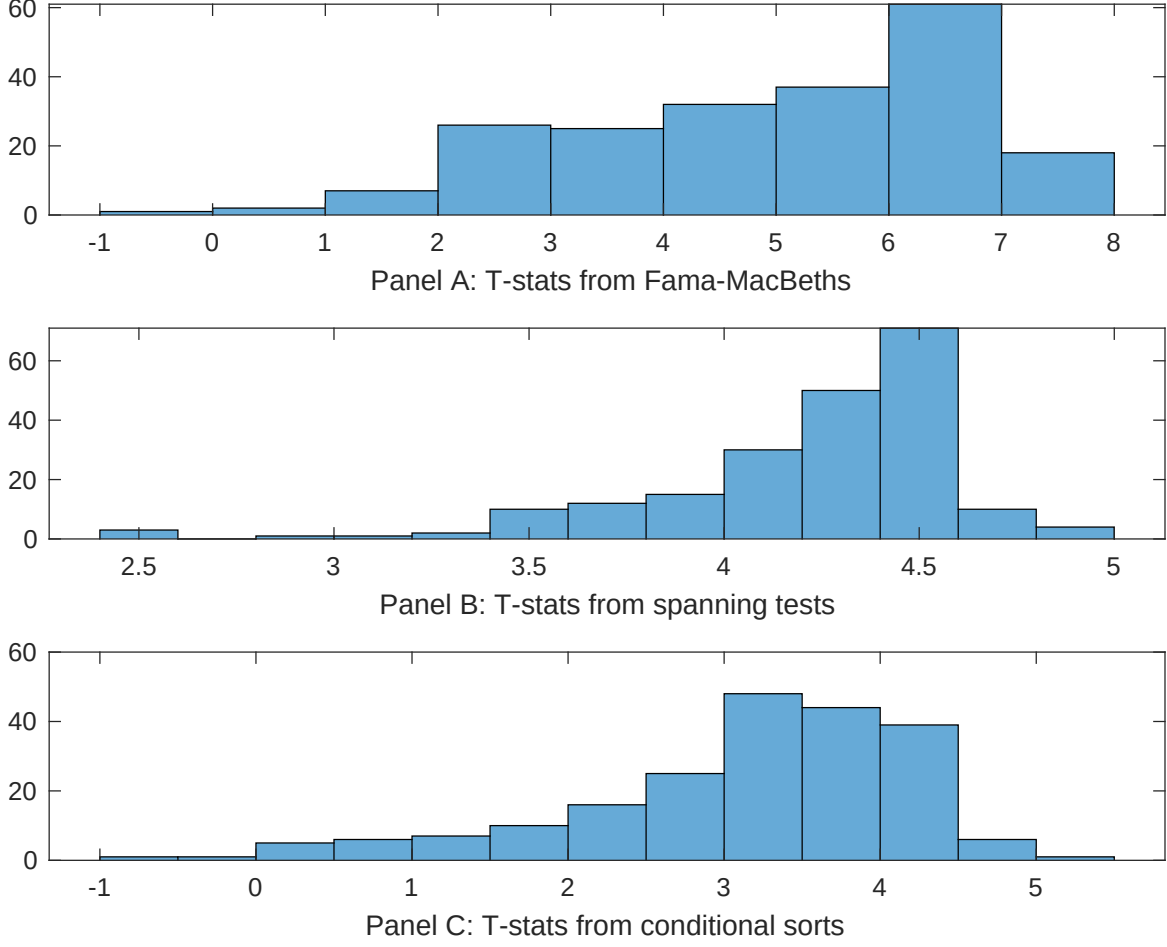


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CIA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CIA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CIA}CIA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CIA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CIA. Stocks are finally grouped into five CIA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CIA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CIA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CIA}CIA_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.14 [5.69]	0.13 [5.33]	0.14 [5.78]	0.14 [5.74]	0.14 [5.66]	0.15 [6.02]
CIA	0.65 [2.53]	0.93 [3.38]	0.76 [2.70]	0.65 [2.33]	0.94 [3.54]	0.57 [2.10]	0.34 [1.20]
Anomaly 1	0.17 [8.90]						-0.63 [-1.53]
Anomaly 2		0.20 [6.45]					0.84 [1.60]
Anomaly 3			0.20 [8.97]				0.64 [1.13]
Anomaly 4				0.10 [8.51]			0.39 [2.01]
Anomaly 5					0.17 [7.53]		0.58 [2.43]
Anomaly 6						0.14 [9.77]	0.57 [3.39]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CIA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CIA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [3.08]	0.21 [3.10]	0.21 [3.15]	0.22 [3.26]	0.22 [3.29]	0.22 [3.23]	0.21 [3.13]
Anomaly 1	13.41 [3.53]						10.93 [2.08]
Anomaly 2		10.68 [3.17]					6.85 [1.86]
Anomaly 3			13.60 [3.72]				5.32 [1.06]
Anomaly 4				7.16 [1.83]			5.35 [1.27]
Anomaly 5					5.60 [1.84]		4.38 [1.33]
Anomaly 6						1.36 [0.35]	-8.19 [-1.82]
mkt	-2.35 [-1.53]	-1.20 [-0.74]	-2.64 [-1.71]	-2.67 [-1.72]	-2.74 [-1.77]	-2.67 [-1.71]	-1.55 [-0.96]
smb	-0.65 [-0.27]	4.15 [1.62]	-0.48 [-0.20]	0.21 [0.09]	0.75 [0.32]	0.75 [0.32]	0.82 [0.30]
hml	-6.89 [-2.31]	-6.59 [-2.20]	-7.23 [-2.43]	-8.11 [-2.71]	-8.38 [-2.78]	-7.88 [-2.61]	-6.11 [-2.01]
rmw	5.31 [1.75]	0.14 [0.04]	5.02 [1.65]	6.54 [2.15]	6.62 [2.18]	6.54 [2.15]	0.82 [0.22]
cma	22.79 [4.86]	20.31 [4.05]	23.74 [5.16]	18.61 [2.84]	22.83 [4.44]	26.25 [4.89]	13.87 [2.05]
umd	2.57 [1.64]	3.66 [2.39]	2.86 [1.85]	3.88 [2.51]	3.72 [2.41]	3.70 [2.38]	2.77 [1.77]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	12	13	11	11	11	14

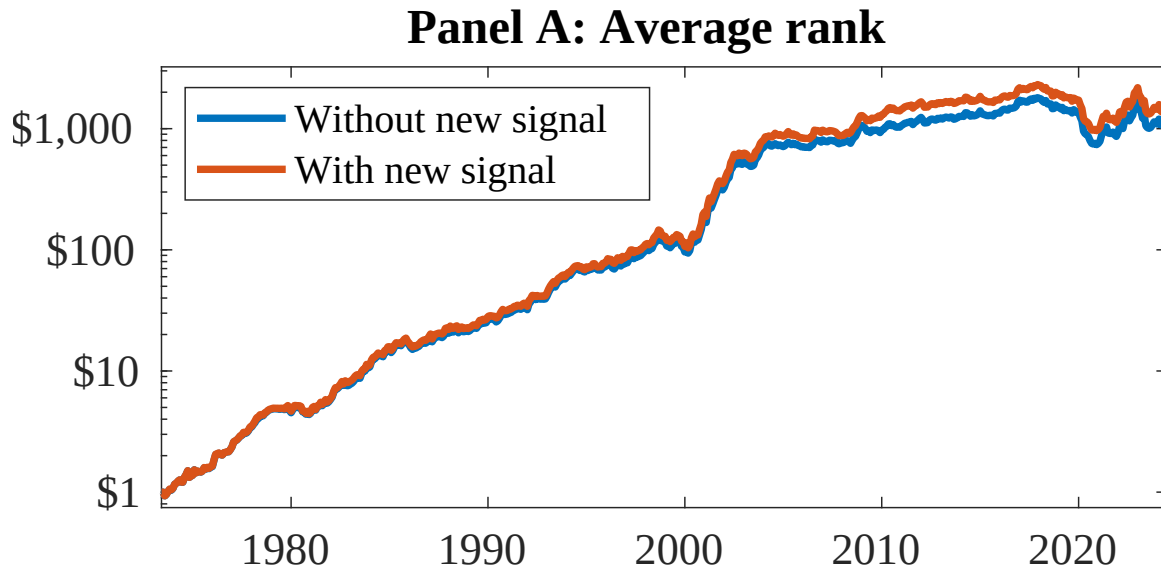


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as CIA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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